

SARVARBEK ERNIYAZOV

Ph.D. Candidate, Computer Engineering | Reliable Learning and Decision-Making for Energy and Cyber-Physical Systems

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[Google Scholar](#) | [GitHub](#) | [Research Portfolio](#) | [Hugging Face](#) | [LinkedIn](#)

Academic Profile

I develop risk-bounded and distribution-resilient learning methods for heterogeneous physical systems. My doctoral research advances spatio-temporal graph learning, multimodal mixture-of-experts, frequency-aware modeling, and federated learning for solar, building-energy, and industrial-control systems; my earlier research in low-power quantum-dot cellular automata established a hardware-aware perspective on energy, latency, and computational constraints. This progression defines my emerging independent direction: determining when a distributed model should use a prediction, defer the decision, or request additional information as sensor and site conditions change.

Research Focus

Spatio-temporal and multimodal learning for energy and cyber-physical systems • Federated and personalized learning under client heterogeneity • Uncertainty quantification, conformal risk control, and selective prediction • Distribution shift, anomaly detection, and failure-centered evaluation • Resource-aware AI for edge–cloud physical infrastructure.

Education and Availability

Ph.D. in Computer Engineering, Chonnam National University, Yeosu, South Korea 2020 – Feb. 2027 (exp.)

Advisor: Prof. Chang Gyoon Lim • Dissertation defense scheduled: **December 2026** • Official degree conferral: **February 2027**

M.S. in Computer Engineering, Kumoh National Institute of Technology, Gumi, South Korea 2016 – 2018

Information Security Laboratory (QCA Research Group)

B.S. in Computer Engineering (Information Security), Tashkent University of Information Technologies, Uzbekistan 2011 – 2015

A continuous information-security thread runs through all three degrees: undergraduate specialization, master’s research in an information-security laboratory, and doctoral work on industrial-control-system security.

Postdoctoral availability: from March 2027, and compatible with later start dates. My university affiliation and grant responsibilities conclude upon degree conferral in February 2027; the funded project continues to 2028, but my personal participation ends at conferral.

Selected Research Impact

- **3 peer-reviewed AI journal articles** (2024–2026), all first-author; **9 peer-reviewed publications** in total with verified records (7 journal, 2 proceedings); one manuscript under review, listed separately.
- **At least 113 citations** across the full record (Google Scholar, September 2026); **88 citations** to the lead-author 2019 *Microelectronic Engineering* article.
- **10 public research repositories** and **7 hosted research interfaces** (code, model releases, reproducibility artifacts, interactive demos).
- Technical contributions to funded Korean national R&D in virtual power plants, edge–cloud energy management, industrial monitoring, and intelligent infrastructure.

Representative Publications

1. **S. Erniyazov** and C. G. Lim. “GNN-enhanced temporal patch segmentation and frequency fusion model for robust solar energy production forecasting.” *Energy Reports*, 13, 4962–4984, 2025. [doi:10.1016/j.egyr.2025.03.063](https://doi.org/10.1016/j.egyr.2025.03.063)
First-author; up to 46% lower reported MSE at 12 h versus PatchTST, Crossformer, PatchMixer, and Fredformer under the study protocol.
2. **S. Erniyazov**, M. A. Jaleel, C. G. Lim, and S. B. Ha. “A multimodal fusion framework for solar forecasting using dynamic GNNs and mixture of experts.” *KSII Transactions on Internet and Information Systems*, 20(3), 1337–1360, 2026. [doi:10.3837/tiis.2026.03.012](https://doi.org/10.3837/tiis.2026.03.012)
First-author; dynamic graph and horizon-aware expert routing; 49% reported forecast skill relative to the NAM physical model.
3. **S. Erniyazov**, Y.-M. Kim, M. A. Jaleel, and C. G. Lim. “Comprehensive analysis and improved techniques for anomaly detection in time series data with autoencoder models.” *International Journal on Advanced Science, Engineering and Information Technology*, 14(6), 2024. [doi:10.18517/ijaseit.14.6.20716](https://doi.org/10.18517/ijaseit.14.6.20716)
First-author; HAI industrial-control-system benchmark; 99% accuracy and $F1 = 0.987$ under the study protocol.
4. **S. Erniyazov** and J.-C. Jeon. “Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation.” *Microelectronic Engineering*, 211, 37–43, 2019. [doi:10.1016/j.mee.2019.03.015](https://doi.org/10.1016/j.mee.2019.03.015) — 88 citations.

Research Experience

Ph.D. Researcher — AI Laboratory, Chonnam National University

2020 – present

Department of Computer Engineering, Yeosu, South Korea • Advisor: Prof. Chang Gyoong Lim

Research program: spatio-temporal, multimodal, and distributed learning for energy forecasting and cyber-physical anomaly detection.

- Led first-author research on graph-enhanced temporal and frequency modeling for multi-station solar forecasting, including architecture design, implementation, baseline comparison, ablation analysis, and manuscript preparation.
- Developed dynamic graph and mixture-of-experts methods for integrating heterogeneous solar and meteorological modalities across forecast horizons.
- Investigated federated spatio-temporal mixture-of-experts for heterogeneous building-energy clients; the manuscript is under review and is excluded from the peer-reviewed publication count.
- Developed and evaluated autoencoder-based time-series anomaly detection on the HAI industrial-control-system benchmark.
- Built reproducible research pipelines: data validation, leakage-aware splitting, training, statistical evaluation, ablations, and research-prototype integration.
- Contributed AI methodology and technical deliverables to Korean national R&D programs involving virtual power plants, edge–cloud energy management, satellite–ground data integration, and power-infrastructure monitoring.

Individual contribution statement: For first-author doctoral studies, I led method development, software implementation, experimental design, baseline and ablation evaluation, result visualization, and manuscript preparation. Formal grant governance and institutional PI responsibilities remained with the appointed faculty principal investigator.

Graduate Researcher — Information Security Laboratory / QCA Research Group

2016 – 2018

Kumoh National Institute of Technology, Gumi, South Korea

- Designed and simulated low-power quantum-dot cellular automata circuits, including full adders, carry-save and carry-look-ahead adders, magnitude comparators, subtractors, and reversible logic.
- Reported more than 50% area reduction for the proposed carry-look-ahead architecture relative to the specified comparison design.
- Contributed to the laboratory's peer-reviewed QCA publication record, including the 2019 *Microelectronic Engineering* article.
- Conducted this work inside an information-security laboratory, extending an undergraduate specialization in information security into hardware-level design. The resulting perspective on energy, latency, sensing, and resource constraints—and on reliability as a security-adjacent property of physical systems—now informs my cyber-physical AI research.

National R&D Technical Leadership

Role formulation used throughout: Lead AI Architect / Lead Researcher for the assigned technical work package; not the formal grant Principal Investigator. I led the assigned AI work package from problem formulation and data-interface design through model architecture, leakage-aware validation, and research-prototype integration, coordinating technical interfaces with the faculty PI, domain partners, and software contributors while grant governance remained with the appointed PI.

TIPA RS-2026-25529976 — sLM–LMM Cooperative Virtual Power Plant Platform

[Funded and active | 2026–2028] Industry–academia commercialization R&D for on-device AI and cloud-integrated energy management. **Technical role:** Lead AI Architect / Lead Researcher for the assigned university AI work package; not the formal grant PI.

- Define model and evaluation architecture for cooperation between on-device small language models and cloud-based multimodal models in VPP workflows.
- Lead AI-side interface design connecting telemetry, forecasts, operator tools, and edge–cloud inference components.
- Direct technical validation for assigned machine-learning deliverables and coordinate research-prototype integration with faculty and industry partners.

Appointment clarification: my university affiliation and associated project responsibilities conclude upon Ph.D. degree conferral in February 2027; no project, funding, or time commitment will continue into a postdoctoral appointment.

TIPA RS-2025-02312851 — Preliminary Study for an On-Device AI and Cloud-Integrated VPP Platform

[Funded and completed | 2025] **Technical role:** Lead AI Researcher for the machine-learning feasibility work: led AI feasibility analysis, model architecture planning, and prototype evaluation supporting the subsequent commercialization-stage project, and translated research requirements into a technically scoped edge–cloud VPP development plan.

Selected Proposal Development — Not Funded Awards

- NRF RS-2025-00564485 [submitted / under review] — satellite–ground data integration for solar variability management and VPP operations; contributed multimodal mixture-of-experts and on-device-AI methodology.
- INNOPOLIS RS-2026-25550513 [submitted / under review] — global validation of AI-based power-equipment anomaly detection and preventive control; contributed AI anomaly-detection and validation methodology.

Submitted proposals are listed separately and are not counted as funded grants.

Publications

Citation counts were verified through Google Scholar in September 2026. Manuscripts under review are separated from published or accepted work and are not included in peer-reviewed publication counts. My name is shown in bold. Full record: [Google Scholar](#) and [ORCID](#).

Record summary 7 peer-reviewed journal articles • 2 peer-reviewed proceedings papers • 1 manuscript under review • first author on every entry

Peer-Reviewed Journal Articles — Published or Accepted

- [J1] **S. Erniyazov**, M. A. Jaleel, C. G. Lim, and S. B. Ha. “A multimodal fusion framework for solar forecasting using dynamic GNNs and mixture of experts.” *KSII Transactions on Internet and Information Systems*, 20(3), 1337–1360, 2026. [doi:10.3837/tiis.2026.03.012](#)
Contribution: first-author multimodal architecture, dynamic graph and expert-routing methodology, evaluation, analysis, and manuscript preparation.
- [J2] **S. Erniyazov** and C. G. Lim. “GNN-enhanced temporal patch segmentation and frequency fusion model for robust solar energy production forecasting.” *Energy Reports*, 13, 4962–4984, 2025. [doi:10.1016/j.egyr.2025.03.063](#)
Contribution: first-author method development, implementation, experimental design, baseline and ablation evaluation, and manuscript preparation.
- [J3] **S. Erniyazov**, Y.-M. Kim, M. A. Jaleel, and C. G. Lim. “Comprehensive analysis and improved techniques for anomaly detection in time series data with autoencoder models.” *International Journal on Advanced Science, Engineering and Information Technology*, 14(6), 2024. [doi:10.18517/ijaseit.14.6.20716](#)
Contribution: first-author model design, HAI benchmark evaluation, analysis, and manuscript preparation.
- [J4] **S. Erniyazov** and J.-C. Jeon. “Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation.” *Microelectronic Engineering*, 211, 37–43, 2019. [doi:10.1016/j.mee.2019.03.015](#)
88 citations (Google Scholar, September 2026). Contribution: circuit design, simulation, comparative evaluation, and manuscript preparation.
- [J5] **S. Erniyazov** and J. C. Jeon. “Coplanar subtractor design in QCA for arithmetic circuit design.” *Advanced Science Letters*, 23(10), 10112–10117, 2017. [doi:10.1166/asl.2017.10399](#)
- [J6] **S. Erniyazov** and J. C. Jeon. “Implementation of area efficient Fredkin gate for compact reversible QCA circuit.” *Advanced Science Letters*, 23(10), 10092–10096, 2017. [doi:10.1166/asl.2017.10395](#)
- [J7] **S. Erniyazov** and J. C. Jeon. “Reversible circuit design in QCA based on double Feynman gate.” *Advanced Science Letters*, 23(10), 9852–9856, 2017. [doi:10.1166/asl.2017.9810](#)

Peer-Reviewed Conference or Proceedings Papers

- [C1] **S. Erniyazov** and J.-C. Jeon. “Area efficient magnitude comparator based on QCA.” *Advanced Science and Technology Letters*, 150, 75–79, 2018. [doi:10.14257/asl.2018.150.19](#)
- [C2] **S. Erniyazov** and J.-C. Jeon. “Half subtractor in QCA for arithmetic circuits.” In *Proceedings of the International Workshop on Future Technology (FUTECH 2017)*, Korean Institute of Information Technology, pp. 148–149, July 2017. [Proceedings record](#)

Manuscripts Under Review — Not Included in Publication Count

- [M1] **S. Erniyazov** et al. “Fed-ST-MoE: A federated spatio-temporal GNN with mixture-of-experts for heterogeneous building energy and demand forecasting.” Manuscript under review.
Investigates decentralized forecasting across heterogeneous buildings with spatio-temporal graph learning and client-aware mixture-of-experts. The study addresses statistical heterogeneity and local-data retention; it does not claim a formal privacy guarantee. Venue and preprint link will be disclosed as permitted by the review policy.

Exploratory Studies in Progress — External Methodological Collaboration, Not Publications

Thermal and audio sensing serve as methodological stress-test environments for my research on distribution-resilient physical AI. Both studies are in progress; blind or external evaluation remains pending, and no peer-reviewed output is claimed. Collaborating institutions and faculty will be named once public attribution is agreed.

- **ThermalGuard** — exploratory study of identity-minimizing thermal perception for fall-event detection under subject, site, and sensor shift; evaluates event-level sensitivity, false alarms, detection delay, and edge-inference constraints. No formal privacy guarantee is claimed.
- **AudioShield** — exploratory cross-corpus anti-spoofing study using WavLM and XLS-R under codec, resampling, replay, and corpus shift; six controlled experiments and a domain-reliance analysis are public. Blind evaluation pending; no final performance or state-of-the-art claim. [Code](#)

Selected Open Research Artifacts

Public methodological studies, reproducibility instruments, and research prototypes; none is a peer-reviewed publication or an operational deployment. Index: sarvarbek-erniyazov.github.io.

WaveSeg — Frequency-Boundary Adapter for Medical Image Segmentation

Public code, model weights, and interactive demo. Adds 770 parameters to SegFormer-B0; reported LGG HD95 decreases from 29.70 to 22.83 pixels; 58.27 FPS at 256×256 on an RTX 4060. Includes patient-level splitting, boundary metrics, ablations, and a documented negative auxiliary-loss result. A methodological artifact, not a claim of clinical validation.

[Code](#) • [Model weights](#) • [Interactive demo](#)

CellTriage — Risk-Controlled Battery Quality Decisions

Public code and interactive research console for early-cycle life prediction, conformal intervals, and cost-aware accept/reject/defer decisions (Extra Trees: MAPE 9.57% ± 3.35, $\rho = 0.944$; 14.2× cost reduction versus AQL sampling, conditional on the stated cost matrix). Reported conformal coverage falls from about 90% in distribution to 60% under held-out campaign shift, documenting failure of nominal coverage when exchangeability is violated.

[Code](#) • [Research console](#)

DriftSentinel — Self-Auditing Distribution-Shift Monitoring

Executable comparison of PSI, KS and chi-square tests with BH-FDR correction, CUSUM, MMD, and BBSD on a 101,766-encounter clinical model. Uses 20-seed negative controls to identify false drift alarms and invalid detector behavior; overturns an initial “8/8 critical” result. Includes 93 automated tests, continuous integration, and determinism checks.

[Code](#) • [Hosted monitoring console](#)

FreqFD — Cross-Generator Deepfake Detection

Public dual-stream RGB and frequency-domain detector (19.4M parameters) evaluated under a leakage-audited cross-generator protocol. Reported unseen-generator AUC improves from 0.588 to 0.686; the remaining generalization limitation is explicitly acknowledged.

[Code](#)

Additional public artifacts: ReproLab (AI4I-2020 reproducibility audit; F1 ceiling 0.9558 derived three ways, spread 0.0015); DALM-DETR (density-adaptive matching, public checkpoints; strongest reported zero-shot transfer, no in-domain superiority claimed); OCRipple (OCR-error propagation study; ANLS 0.848, layout mAP@50 73.8%); FinRiskGuard; Auto Damage Segmentation; Uzbek Handwriting OCR (TrOCR model release).

Funding Register

Program / ID	Project and role	Status
TIPA RS-2026-25529976	sLM-LMM Cooperative VPP Platform for on-device AI and cloud-integrated energy management. Lead AI Architect / Lead Researcher, assigned university AI work package; not grant PI.	Funded, active, 2026–2028
TIPA RS-2025-02312851	Preliminary study for an on-device AI and cloud-integrated VPP platform. Lead AI Researcher for ML feasibility, architecture planning, and prototype evaluation.	Funded, completed, 2025
NRF RS-2025-00564485	Satellite-ground data integration for solar variability management and VPP operations. Methodology contributor.	Submitted proposal
INNOPOLIS RS-2026-25550513	Global validation of AI-based power-equipment anomaly detection and preventive control. Methodology contributor.	Submitted proposal
Earlier programs	NRF/MSIP NRF-2017R1D1A3B03034346; MOE Regional Innovation Strategy 2021RIS-002; MSS RS-2023-00278623. Research contributor.	Funded, completed

Methods

Spatio-temporal graph learning; multimodal fusion; mixture-of-experts; federated and personalized learning; time-series forecasting; anomaly and drift detection; conformal prediction; uncertainty quantification; selective prediction; computer vision; frequency-domain learning; and edge-aware evaluation.

Open Science and Academic Service

- Maintainer of public reproducibility artifacts, model cards, and research demonstrations supporting transparent evaluation (leakage audits, negative controls, multi-seed reporting, automated tests, and continuous integration across the artifacts listed above).
- Laboratory service at the AI Laboratory, Chonnam National University: research-pipeline and evaluation-protocol support for national R&D deliverables.